W003 + W004 Combined Build Report — Layer Hierarchy & Geometric Tracing

Project: MINERVA CryoCell / QCELL & RFCELL P&ID — colour-line-first model
Programme: Mott MacDonald / SCK CEN / MYRRHA-MINERVA Phase 1
Waves: W003 (Layer Hierarchy) + W004 (Geometric Tracing) — combined 10-phase build
Baseline: W002 (`5d43a9e`)
Toolchain: Python standard library + `cairosvg` (PDF render) + `openpyxl` (Excel)

Executive summary

This build extends the W002 colour-line decomposition with a full hierarchy of **13 top-level layers (21 named sub-layers)** and a geometric flow-tracing pass. Headline outcomes:

- Unmapped elements reduced **982 → 112** (88.6 % resolved).
- **13 top-level layers (21 named sub-layers)** assigned; QCELL annotated **1834** elements (14 sub-layers used), RFCELL **591** (12 sub-layers).
- **132** flow arrows traced, **77** floating arrows isolated, **36** junctions.
- **297** components cataloged to Excel + HTML.
- **22** handover diamonds and **5/5** scope boundaries validated.
- PEMO SSOT emitted as YAML 1.2 with **122** control loops and **60** heat loads.

The build follows “Claim ≠ Complete” discipline: every number below is a real runtime count from the engines. Items that cannot yet be computed are marked **DEFERRED** and tracked in `docs/TODO.md` and `docs/CAPABILITY_MATRIX.md`.

Phase 1 — Legend-based unmapped reduction

Metric	Count
Unmapped before	982
Reclassified	870
Still unresolved	112 (UNRESOLVED_OTHER_COLOUR / magenta family)

Top categories: symbol/heatload markers 236, leader/signal lines 211, uncoloured spec-dots 153, instrument bubbles 146, structure paths 57, flow arrows 40, boundary/titleblock 27. Detail → `reports/unmapped_reduction_analysis.md`.

Phase 2 — Element pairing engine

Pairing	Count
Text → component pairs	315
Text colour-classified	533
Dots paired to lines	205
Heat-load triangles paired	100
Arrows paired to lines	132
Arrows floating (unpaired)	77

Pairing-distance quality (752 paired elements with `distance_px`): median **25.35 px**, p90 **355.4 px**, max **1040.2 px**. On a 1527 px-wide canvas a 1040 px pairing is almost certainly wrong, so pairs above the p90 (~355 px) are flagged **LOW-CONFIDENCE** and must not be trusted as ground truth; the **77** floating arrows are left unmatched rather than force-paired (governance: no silent deletion of unresolved data). Tight median (25 px) confirms the bulk of pairings are sound.

Output → `data/model/paired_elements.json`.

Phase 3 — Text standardization

Target font `Consolas, 'DejaVu Sans Mono', monospace`; 4 mm-based tiers.

Tier	mm	Nodes
major_header	3.0	148
instrument_tag	2.2	157
segment_label_vertical	2.5	17
annotation	1.8	211

533 text nodes total. Detail → `reports/text_standardization_report.md`.

Phase 4 — 13 top-level layers (21 named sub-layers)

Per-layer element counts (assignment model):

Layer	Count
02_ScopeBoundaries_Main	565
04A_Lines_A_BLUE	103
04B_Lines_B_CYAN	43
04C_Lines_W_GREEN	61
04D_Lines_S_OLIVE	67
04E_Lines_V_GREY	40
04F_Lines_D_ORANGE	65
05_HeatLoads_ALL	100
06_SegmentNames_Vertical_Black	17
07_Equipment_Major	372
08_Instruments_Bubbles	162
09_Control_Elements	51
10_Signals_Dashed	54
11_Text_ColorCoded	516
12_Dots_SpecChanges_ALL	205

Empty layers (00 grid, 01 titleblock, 03A/03B manifolds, 04G_E_RED, 13 legend) are reserved placeholders for future waves. Rendered artifacts:

- `output_v6/QCELL/QCELL_13layers.svg` + `.pdf` — 1834 elements annotated, 14 layers used
- `output_v6/RFCELL/RFCELL_13layers.svg` + `.pdf` — 591 elements annotated, 12 layers used
- Interactive layer-toggle atlas → `publish/layered_atlas_v6.html`

Elements are annotated **in place** with `class="lyr-NN"` to preserve CTM transforms.

Layer sum-check (exact reconciliation): `1888 drawable + 533 text = 2421` total elements assigned, with **no drops and no double-counts** — every input element lands in exactly one layer bucket. (The 1888 drawable = 1375 QCELL + 513 RFCELL; shapes: 670 triangle / 374 line / 239 path / 205 dot / 162 bubble / 132 arrow / 106 rect.) The in-place atlas annotation reports **2425** (1834 QCELL + 591 RFCELL); the **+4** difference versus 2421 is group-wrapper `<g>` elements that receive a class during annotation but are not leaf drawables — documented here honestly rather than reconciled away.

Phase 5 — Geometric tracing (flow topology)

Metric	Count
Flow arrows traced	132
Floating arrows	77
Junctions	36

Output → `data/model/flow_topology.json`.

Phase 6 — Vertical-letter nomenclature parser

Parsed **33** segment labels into a parent-header tree:

Parent	Children
A (4.5 K main, BLUE)	A, A’, AK, AK1, AK2, AL, AL1, AL2, AS, AX, AZ1, AZ2
D (manifold, ORANGE)	D, D’, DJ, DS, DZ
V (vent, GREY)	V501, V502, V503, VH
E (manifold, RED)	E, E’
B (2 K, CYAN)	B, B’, BZ
S (warm, OLIVE)	S, SW
W (coupler, GREEN)	W, W’
U (TOP-LEFT)	**DEFERRED** — currently black, flagged for recovery

Output → `data/model/segment_nomenclature.json`.

Phase 7 — Component catalog

297 components cataloged with line assignment, tag, prefix and coordinates.
Spec-change dots by line: S 40, V 10, B 2, uncoloured 153.

- `reports/COMPONENT_CATALOG.xlsx`
- `publish/component_catalog.html`
- `data/model/spec_dots_catalog.json`

Phase 8 — Scope boundary validation

5/5 boundaries (QM, QVB, vacuum_barrier, Jumper, QINFRA), **22** handover diamonds (TP#101…TP#604), **19** W-line bottom-right elements. Detail → `reports/scope_boundary_validation.md`.

Phase 9 — PEMO YAML 1.2 SSOT

`data/pemo/ic_system_v1.2.yaml` — **122** control loops, **60** heat loads emitted as a single source of truth.

Phase 10 — Documentation

`docs/VERSION_CHANGELOG.md`, `docs/BUILD_MANIFEST.json`, `docs/CAPABILITY_MATRIX.md`, `docs/TODO.md`, `docs/GITHUB_PR_PLAN.md`, plus this report and the layered atlas.

Deferred / honest gaps

Item	Status
112 magenta / non-canonical colour elements	DEFERRED — no legend swatch
U line (top-left, currently black)	DEFERRED — recovery in future wave
04G_E_RED line layer (0 elements)	DEFERRED — RED currently maps to E text only
77 floating arrows	Isolated, not yet bound to a source line
Manifold COLD/WARM header layers (03A/03B)	Reserved placeholders

Reproduction

```
cd /home/ubuntu/pid_project
PYTHONPATH=src python3 -m abacus_svg_pid.build_w003_w004   # phases 1-9 models
PYTHONPATH=src python3 -m abacus_svg_pid.build_catalog     # phase 7 xlsx+html
PYTHONPATH=src python3 -m abacus_svg_pid.build_atlas_v6    # phase 4 svg/pdf/atlas
```

All HTML deliverables are static files (no server required).